

CHANNEL LIGHTSHIP, United Kingdom

WMO: **995920**

Lat: **49.900N** Lon: **2.900W** Elev: **0** StdP: **14.70** Time Zone: **0.00 (EUW)** Period: **94-18** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	36.6	38.6	24.7	18.8	38.8	27.1	21.1	40.2	50.8	47.3	47.5	47.3	26.1	30	0.870	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	3.1	65.2	62.5	64.1	61.7	63.4	61.0	63.3	64.5	62.5	63.6	61.7	62.9	13.3	230	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	62.8	85.4	64.1	61.8	82.6	63.3	61.0	80.2	62.6	28.7	64.7	28.1	63.7	27.6	63.1	72.7	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	44.9	40.8	36.3	DB	34.4	68.4	2.9	2.9	32.3	70.5	30.6	72.2	28.9	73.8	26.8	75.8	(4)
(5)				WB	30.9	64.9	2.2	2.4	29.3	66.7	28.0	68.1	26.8	69.4	25.2	71.2	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		53.5	47.6	46.6	47.4	49.3	52.5	56.5	59.9	61.3	60.2	57.5	53.2	49.4	(6)
(7)		DBStd		6.00	4.10	3.82	3.05	2.69	2.37	2.07	1.96	1.59	1.95	2.85	3.48	4.21	(7)
(8)		HDD50		400	91	100	85	42	6	0	0	0	0	1	13	62	(8)
(9)		HDD65		4197	539	515	545	471	386	254	158	114	143	234	354	484	(9)
(10)		CDD50		1677	17	5	5	21	85	196	307	351	307	232	109	42	(10)
(11)		CDD65		1	0	0	0	0	0	0	0	1	0	0	0	0	(11)
(12)		CDH74		0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80		0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg		19.3	24.3	22.5	19.9	17.1	16.5	14.6	14.7	14.8	17.9	21.5	23.4	24.6	(14)
(15)	Precipitation	PrecAvg		33.0	3.7	2.9	2.2	2.1	2.0	1.8	1.7	2.2	2.2	3.6	4.4	4.2	(15)
(16)		PrecMax		40.6	7.7	5.7	6.5	5.3	4.5	4.1	3.2	5.4	5.7	6.8	8.8	9.6	(16)
(17)		PrecMin		25.0	0.5	0.5	0.2	0.0	0.4	0.2	0.4	0.4	0.2	0.6	1.1	0.8	(17)
(18)		PrecStd		4.4	1.9	1.5	1.3	1.3	1.2	1.0	0.8	1.2	1.2	1.6	1.9	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		55.7	54.8	53.8	55.7	60.0	63.5	66.5	67.4	66.0	64.2	60.4	57.6	(19)
(20)			MCWB		53.0	51.4	50.6	52.4	55.7	60.0	63.4	64.6	62.3	61.4	57.8	54.7	(20)
(21)		2%	DB		54.6	53.0	52.2	54.0	57.4	61.3	64.7	65.5	64.3	62.8	59.2	56.7	(21)
(22)			MCWB		52.2	50.3	49.5	51.3	54.6	58.6	62.2	63.0	61.9	60.4	56.7	54.2	(22)
(23)		5%	DB		53.7	52.0	51.5	53.1	56.2	60.1	63.5	64.4	63.5	61.9	58.4	55.8	(23)
(24)			MCWB		51.2	49.4	49.0	50.6	53.8	57.6	61.2	61.9	61.0	59.5	55.8	53.2	(24)
(25)		10%	DB		52.7	51.1	50.9	52.2	55.2	59.2	62.6	63.6	62.8	61.0	57.6	54.9	(25)
(26)			MCWB		50.1	48.5	48.4	49.9	53.1	56.9	60.3	61.2	60.3	58.6	54.9	52.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		54.0	52.1	51.7	53.3	56.7	61.1	64.3	65.2	63.6	62.2	58.5	55.8	(27)
(28)			MCDB		54.9	54.1	52.7	54.7	58.4	62.8	65.4	66.9	64.5	63.2	59.9	56.8	(28)
(29)		2%	WB		52.8	51.0	50.4	52.0	55.4	59.4	62.8	63.7	62.6	61.1	57.4	54.8	(29)
(30)			MCDB		54.1	52.6	51.6	53.4	56.7	60.7	64.1	65.0	63.7	62.3	58.9	56.2	(30)
(31)		5%	WB		51.7	50.0	49.7	51.2	54.5	58.3	61.7	62.7	61.8	60.2	56.4	53.9	(31)
(32)			MCDB		53.1	51.6	51.1	52.6	55.7	59.5	63.0	63.8	63.0	61.5	58.2	55.5	(32)
(33)		10%	WB		50.6	49.1	48.9	50.4	53.7	57.3	60.9	61.9	61.0	59.3	55.4	52.7	(33)
(34)			MCDB		52.5	50.9	50.6	51.9	54.9	58.6	62.1	63.1	62.4	60.8	57.4	54.7	(34)
(35)	Mean Daily Temperature Range		MDBR		4.4	4.1	3.6	3.6	3.4	3.5	3.2	3.1	3.3	3.6	3.9	4.5	(35)
(36)		5% DB	MCDBR		4.7	4.5	3.8	4.2	4.5	5.0	4.5	4.1	4.1	4.0	3.9	4.7	(36)
(37)			MCWBR		5.8	5.6	4.2	3.7	3.8	4.1	3.9	3.9	3.9	4.3	5.0	6.1	(37)
(38)		5% WB	MCDBR		4.6	4.5	3.8	4.0	4.2	4.6	4.3	4.0	4.0	3.7	4.0	4.9	(38)
(39)			MCWBR		6.1	5.7	4.5	3.7	3.9	4.2	4.0	4.0	4.0	4.3	5.3	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.339	0.354	0.388	0.399	0.403	0.396	0.391	0.383	0.375	0.366	0.339	0.331	(40)
(41)		taud			2.382	2.386	2.308	2.308	2.352	2.404	2.421	2.462	2.464	2.450	2.431	2.352	(41)
(42)		Ebn at Noon			219	247	258	268	272	274	273	271	261	240	219	202	(42)
(43)		Edh at Noon			21	27	34	37	37	36	35	32	29	25	20	19	(43)
(44)	All-Sky Solar Radiation	RadAvg			268	509	903	1450	1750	1901	1827	1527	1158	650	339	217	(44)
(45)		RadStd			22	60	103	121	113	186	109	118	77	45	31	25	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page